# BREAK-EVEN ANALYSIS, Part-2

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# Important Formulae

$$\text{Break-even point (in units)} = \frac{\text{Fixed cost}}{\text{Contribution per unit}}$$

$$\text{Break-even point (in value)} = \frac{\text{Fixed cost}}{\text{Contribution per unit}} \times \text{Selling price per unit}$$

$$= \frac{\text{Fixed cost}}{\text{PV Ratio}}$$

# Contribution

Contribution= Sales–Variable Cost

Contribution=Fixed Cost + Profit

As long as the contribution is more than the fixed cost, there is profit.

Contribution = Sales × PV Ratio

Sales–Variable Cost=Sales× PV Ratio

Sales =Variable Cost/(1–PV ratio)

# Fixed Cost

Fixed Cost=Contribution $-$ Profit

$=$ (Sales $\times$ PV ratio) $-$ Profit

If PV Ratio is 1, then

Fixed Cost= Sales $-$ Profit

$\Rightarrow$Profit= Sales $-$ Fixed Cost

$\Rightarrow$Variable cost is zero.

If PV Ratio is 0, Fixed Cost= $-$ Profit=Loss

# Profit

Profit =(Sales × PV)–Fixed Cost

$$\text{Sales (in units)} = \frac{\text{Fixed cost} + \text{Profit}}{\text{Contribution per unit}}$$

$$\text{Sales (in value)} = \frac{\text{Fixed cost} + \text{Profit}}{\text{PV Ratio}}$$

# Profit-Volume Ratio

- Also called Contribution-to-Sales Ratio

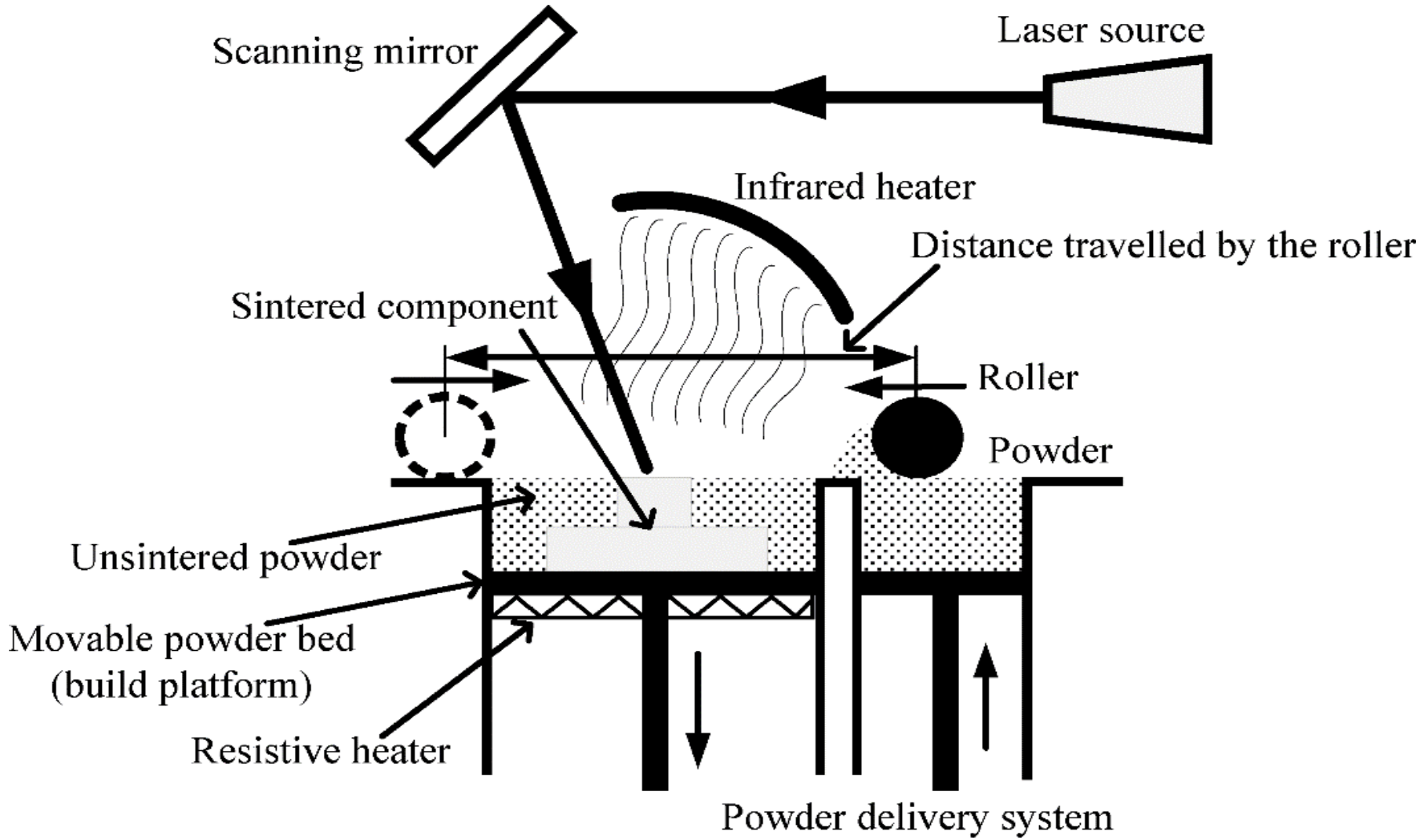
$$\text{PV Ratio} = \frac{\text{Contribution}}{\text{Sales}}$$

$$\%\text{PV Ratio} = \frac{\text{Contribution}}{\text{Sales}} \times 100$$

$$= \frac{\text{Change in Profit}}{\text{Change in Sales}} \times 100$$

- PV ratio may differ from product to product.

# A case study of using a Selective Laser Sintering



# Proper utilization of SLS machine

- **More units we make in a day, more profit.**
- **Hence, the machine should be utilized properly.**
- **We have to any way pay for fixed cost (depreciation, maintenance, supervision etc.)**
- **Hence, better not to keep it under-utilized**

Sharma, F., Dixit, U.S. An Analytical Method for Assessing the Utility of Additive Manufacturing in an Organization. *J. Inst. Eng. India Ser. C* **102**, 41–50 (2021). https://doi.org/10.1007/s40032-020-00624-0

# Utility Assessment

## Evaluation of Utilization Factor $U_f$

$$U_f = \frac{\text{Time required for production}}{\text{Total available time}}$$

- If $U_f$ is less than 1, the machine is under-utilized. Management should plan to get job orders from outside.
- If $U_f$ is more than 1, the machine is over-utilized. Management may have to outsource the jobs or plan to procure additional machine.
- An organization may have a policy to run only one shift in a day and may have some holidays in a year (6 hours per day, 300 days in a year). In that case, if $U_f$ is more than 1 but less than a threshold value, the products can be manufactured on the machine with provision for overtime.

$$U_{f\max} = \frac{365 \times 24}{300 \times 6} = 4.87$$

## Impact of Utilization Factor on Cost

For $U_f$=1, i.e., for full utilization, a fundamental cost formula for 3D Printing can be written as

$$C_{total} = \left(c_{material} m_r\right) + \left(\frac{c_{slicing} t_{design}}{n_p}\right) + \left\{\left(c_{operator} + c_{overhead}\right) T_{build}\right\} + \left(c_{operator} t_{pp}\right).$$

$m_r$ is the mass of raw material needed for fabricating the part. $t_{pp}$ is the time required to remove the part from the build chamber of the machine and perform finishing operations

Utilization factor will affect the overhead cost in an inverse manner; lesser the $U_f$, the more will be $C_{total}$, and vice versa.

$$C_{total} = \frac{c_{overhead} T_{build}}{U_f}\left(1 + \frac{U_f C_{material}}{c_{overhead} T_{build}} + \frac{U_f c_{operator}}{c_{overhead}} + \frac{U_f c_{slicing} T_{design}}{n c_{overhead} T_{build}} + \frac{U_f C_{post\text{-}processing}}{c_{overhead} T_{build}}\right).$$

Ratio $r$ of actual cost to the cost with $U_f$=1 is given by

$$r = \frac{\frac{1}{U_f}\left(1 + \frac{U_f C_{material}}{c_{overhead} T_{build}} + \frac{U_f c_{operator}}{c_{overhead}} + \frac{U_f c_{slicing} T_{design}}{n c_{overhead} T_{build}} + \frac{U_f C_{post\text{-}processing}}{c_{overhead} T_{build}}\right)}{\left(1 + \frac{C_{material}}{c_{overhead} T_{build}} + \frac{c_{operator}}{c_{overhead}} + \frac{c_{slicing} T_{design}}{n c_{overhead} T_{build}} + \frac{C_{post\text{-}processing}}{c_{overhead} T_{build}}\right)}.$$

# Incorporation of Uncertainty in the Analysis

The total available time can also be represented as a triplet $(t_{al}, t_{am}, t_{au})$

Similarly, the time required for production can be represented as $(t_{pl}, t_{pm}, t_{pu})$

The utility factor can be computed as $$U_f = \frac{(t_{pl}, t_{pm}, t_{pu})}{(t_{al}, t_{am}, t_{au})} = \left( \frac{t_{pl}}{t_{au}}, \frac{t_{pm}}{t_{am}}, \frac{t_{pu}}{t_{al}} \right)$$

To assist the decision process about the utilization of 3D printing machine, the fuzzy number $U_f$ is subtracted by 1 and is denoted by an index $\beta$.

$$\beta = U_f - 1 = \left( \frac{t_{pl}}{t_{au}} - 1, \frac{t_{pm}}{t_{am}} - 1, \frac{t_{pu}}{t_{al}} - 1 \right)$$

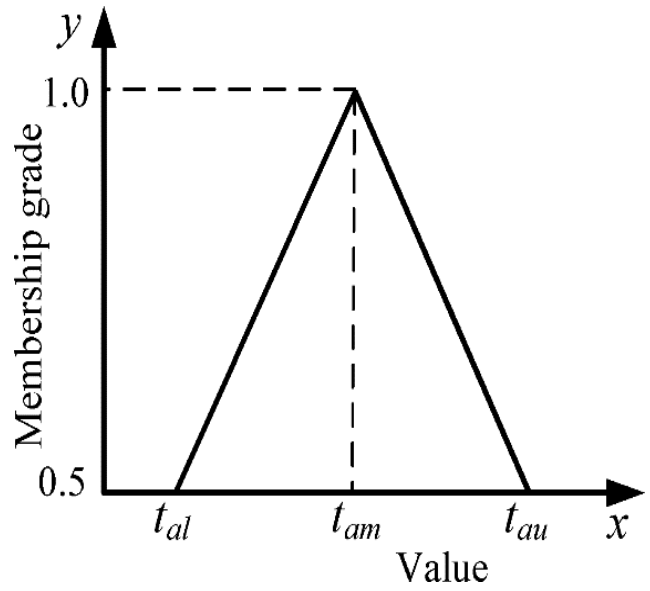



**Figure** A triangular membership function of total available time

The decision strategy can be as follows

$$\beta = U_f - 1 = \left( \frac{t_{pl}}{t_{au}} - 1, \frac{t_{pm}}{t_{am}} - 1, \frac{t_{pu}}{t_{al}} - 1 \right)$$

- If all the elements of the triplet $\beta$ are negative, then the machine is underutilized. In that case, try to get job orders from outside.
- If all the elements of the triplet $\beta$ are positive, then the machine is over-utilized. Possibility of overtime, outsourcing and procuring another machine should be considered.
- Case of some elements of $\beta$ being positive and some negative needs careful consideration. Here, at some value of $\alpha$-cut, the machine is underutilized and at others it is over-utilized.
- The decision is dependent on how much risk management wants to take. In a simplistic way, the centroid of $\beta$ can be evaluated. The machine is underutilized for negative value of centroid and vice versa.

$$r = \frac{\frac{1}{U_f}\left(1 + \frac{U_f C_{material}}{c_{overhead} T_{build}} + \frac{U_f c_{operator}}{c_{overhead}} + \frac{U_f c_{slicing} T_{design}}{n c_{overhead} T_{build}} + \frac{U_f C_{post\text{-}processing}}{c_{overhead} T_{build}}\right)}{\left(1 + \frac{C_{material}}{c_{overhead} T_{build}} + \frac{c_{operator}}{c_{overhead}} + \frac{c_{slicing} T_{design}}{n c_{overhead} T_{build}} + \frac{C_{post\text{-}processing}}{c_{overhead} T_{build}}\right)}$$

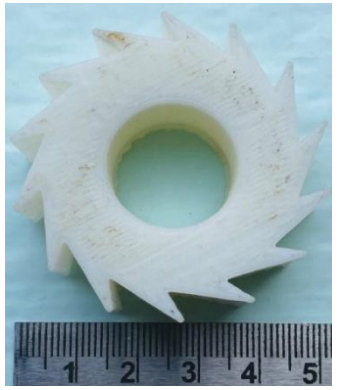
Ratio $r$, and costs can also be computed in the fuzzy form. However, care has to be taken when a particular variable occurs more than once in an expression. In that case, blind application of fuzzy arithmetic will provide wider range.

$$r = \frac{\text{other cost} + \frac{\text{overhead cost}}{U_f}}{\text{other cost} + \text{overhead cost}} = \frac{1 + \frac{x}{U_f}}{1 + x}$$

According to vertex method, expression $r$ has to be evaluated at four combinations of $x$ and $U_f$.

- For $U_f<1$, the lower limit of $r$ is obtained at lower limit of $x$ along with upper limit of $U_f$. For $U_f>1$, the lower limit of $r$ is obtained at upper limit of $x$ along with upper limit of $U_f$.
- For $U_f<1$, the upper limit of $r$ is obtained at upper limit of $x$ along with lower limit of $U_f$. For $U_f>1$, the upper limit of $r$ is obtained at lower limit of $x$ along with lower limit of $U_f$.

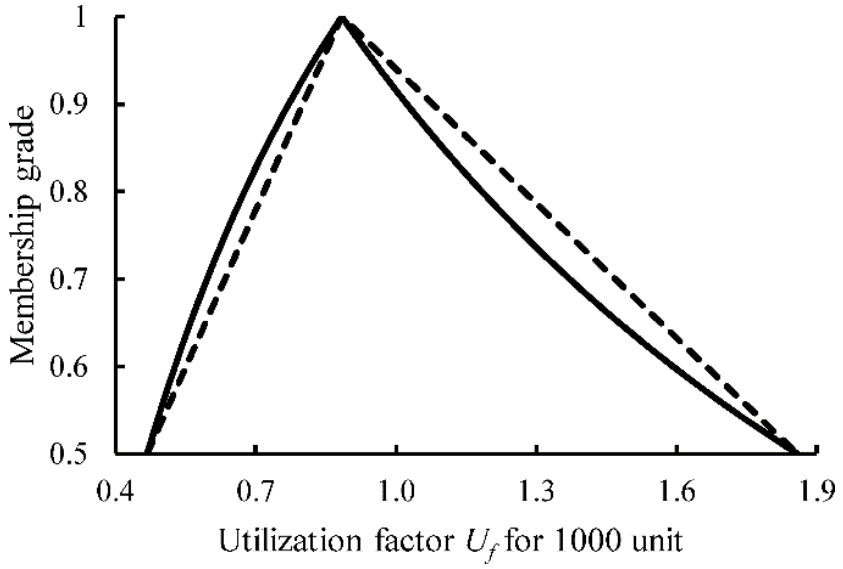
# Case Studies



- The methodologies developed in this work is illustrated by means of some examples.
- Detailed estimation of build time and cost is already discussed.
- First, an example of fabricating a polymer cutter in varying quantities is presented
- Then, an example is taken in which 3D printing machine is required to print two different products (Part A and Part B) in varying quantities
- It is considered that an industry produces a polymer cutter on a SLS machine. Details of geometry of cutter, type of SLS machine, build time and cost are provided
- It is assumed that machine operates for 6 hours daily, 300 days in a year.
- However, the lower estimate for machine operation is only 4 hours due to maintenance etc. and the upper estimation is 9 hours daily, considering some overtime.

**Table** Details of machine utilization and costs for producing different quantities of polymer cutters on an SLS 3D printing machine

| Annual production (unit) | Production time (hr) ($t_{pl}$, $t_{pm}$, $t_{pu}$) | Utilization factor ($U_{fl}$, $U_{fm}$, $U_{fu}$) | $\beta$ ($\beta_l$, $\beta_m$, $\beta_u$) | Unit cost ($) for $U_f$=1 ($C_l$, $C_m$, $C_u$) | Actual unit cost ($) ($C_{al}$, $C_{am}$, $C_{au}$) | $r$ ($r_l$, $r_m$, $r_u$) |
|---|---|---|---|---|---|---|
| 100 | (126, 159, 223) | (0.05, 0.09, 0.19) | (–0.95,–0.91,–0.81) | (19.01, 23.40, 30.12) | (36.89, 83.01, 217.86) | (1.75, 3.55, 8.48) |
| 300 | (378, 477, 669) | (0.14, 0.27, 0.56) | (–0.86, –0.74, –0.44) | (18.99, 23.39, 30.11) | (22.23, 39.40, 86.60) | (1.14, 1.68, 3.25) |
| 600 | (756, 954, 1338) | (0.28, 0.53, 1.12) | (–0.72, –0.47, 0.12) | (18.99, 23.38, 30.10) | (18.57, 28.50, 53.75) | (0.96, 1.22, 1.95) |
| 1000 | (1260, 1590, 2230) | (0.47, 0.88, 1.86) | (–0.53, –0.12, 0.86) | (18.99, 23.38, 30.10) | (17.10, 24.14, 40.61) | (0.83, 1.03, 1.42) |
| 1500 | (1890, 2385, 3345) | (0.70, 1.33, 2.79) | (–0.30, 0.33, 1.79) | (18.99, 23.38, 30.10) | (16.37, 21.96, 34.04) | (0.77, 0.94, 1.16) |
| 2100 | (2646, 3339, 4683) | 0.98, 1.86, 3.90) | (–0.02, 0.86, 2.90) | (18.99, 23.38, 30.10) | (15.95, 20.72, 30.29) | (0.73, 0.89, 1.01) |
| 2620 | (3301, 4166, 5843) | (1.22, 2.31, 4.87) | (0.22, 1.31, 3.87) | (18.99, 23.38, 30.10) | (15.74, 20.10, 28.42) | (0.71, 0.86, 0.97) |



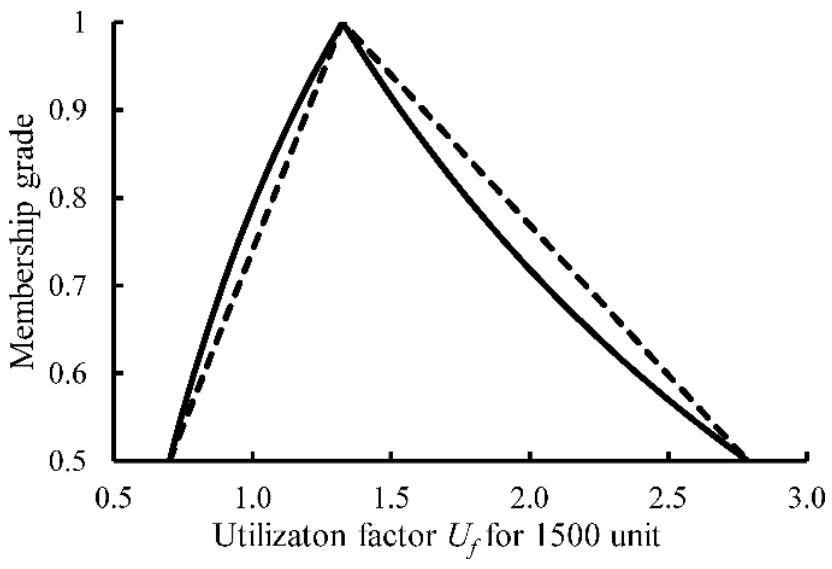
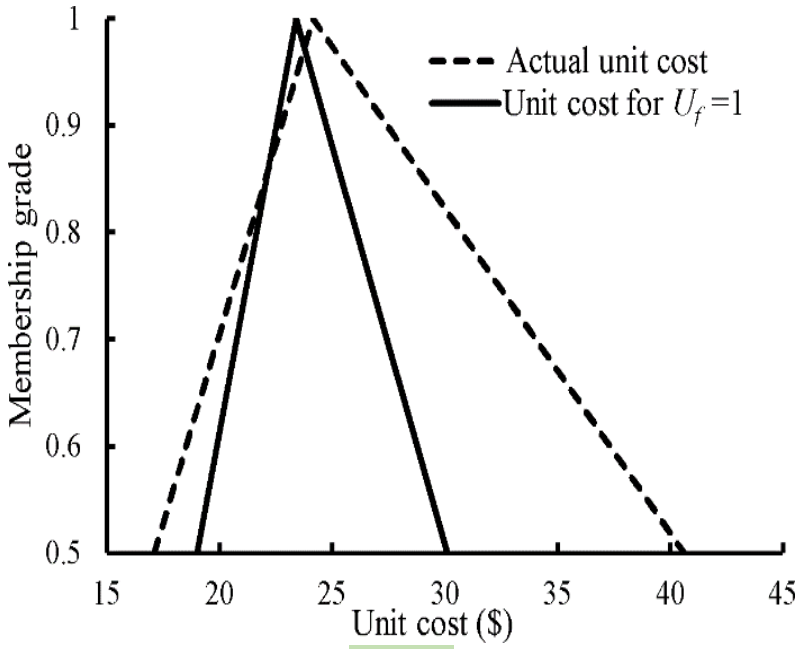
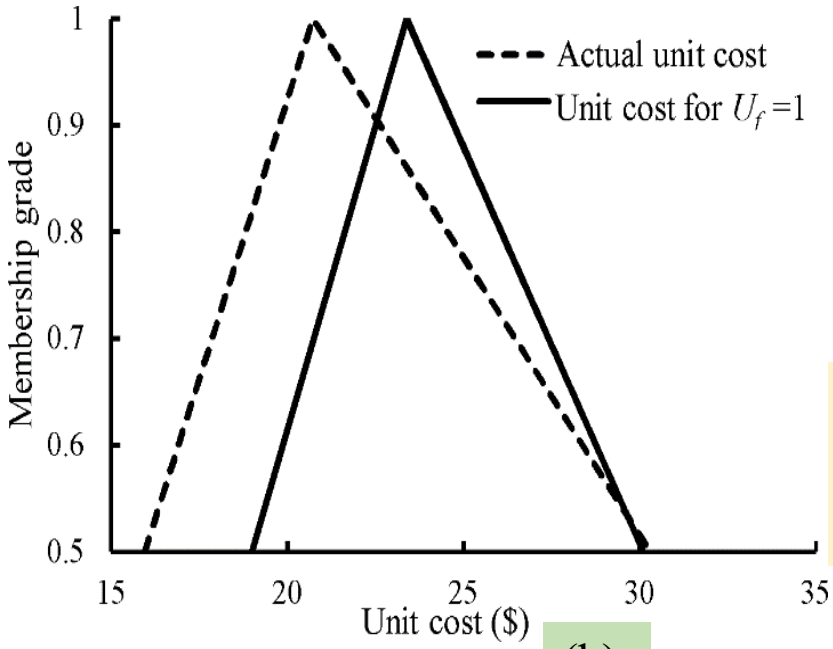




**Figure** Variation of utilization factor $U_f$ with membership grade for the production of 1000, and 1500 units




(a)




(b)

**Figure** Variation of unit cost with membership grade for the production of (a) 1000, and (b) 2100 units

## Case of Producing Two Different Products

**Table** Details of machine utilization and costs for producing different quantities of two parts on an SLS 3D printing machine

| Annual production (unit) | | Production time (hr) ($t_{pl}$, $t_{pm}$, $t_{pu}$) | | Utilization factor ($U_{fl}$, $U_{fm}$, $U_{fu}$) | Actual unit cost ($) ($C_{al}$, $C_{am}$, $C_{au}$) | |
|---|---|---|---|---|---|---|
| Part A | Part B | Part A | Part B | | Part A | Part B |
| 100 | 100 | (126, 159, 223) | (19.97, 26.06, 37.50) | (0.05, 0.10, 0.22) | (33.72, 73.79, 190.92) | (4.89, 11.43, 31.22) |
| 100 | 200 | (126, 159, 223) | (39.95, 52.11, 74.99) | (0.06, 0.12, 0.25) | (31.36, 66.85, 170.46) | (4.50, 10.28, 27.76) |
| 200 | 100 | (252, 318, 446) | (19.97, 26.06, 37.50) | (0.10, 0.19, 0.40) | (25.04, 47.83, 112.24) | (3.52, 7.18, 17.99) |
| 1000 | 1000 | (1260, 1590, 2230) | (194.82, 255.21, 368.65) | (0.54, 1.03, 2.17) | (16.79, 23.24, 37.97) | (2.18, 3.10, 5.42) |

A worth noting fact is that an increase in the quantities of one part reduces the cost of both the parts

- It is seen that for full utilization of the machine, the cost is not dependent on the production volume, but it is dependent on the production volume if it affects utilization.
- Hence, it is stressed that an organization should attempt to enhance the utilization of 3D printing machine.

# Inference

- The increase in the production volume of a part increases the utilization factor with a reduction in the unit cost. The uncertainty in the unit cost for high production volume reduces due to the low impact of overhead cost.
- For the production of two different parts, the increase in the production of one quantity decreases the cost of both the parts. This is due to overall utilization of the machine.

# Importance of non-quantifiable Benefits

- Company’s mission
- Focus
- Likelihood of success
- Market conditions
- Customer goodwill